

Equations of the First Order But Not of The First Degree, Singular Solutions And Extraneous Loci

PART 1: DIFFERENT METHODS OF FINDING GENERAL SOLUTIONS

1.1 Equations of the first order but not of the first degree

Definition. The general first order differential equation of degree $n > 1$ is

$$P_0 (dy/dx)^n + P_1 (dy/dx)^{n-1} + P_2 (dy/dx)^{n-2} + \dots + P_{n-1} (dy/dx) + P_n = 0, \quad \dots (1)$$

where $P_0, P_1, P_2, \dots, P_{n-1}, P_n$ are functions of x and y .

It is a convention to denote dy/dx by p and so (1) becomes

$$P_0 p^n + P_1 p^{n-1} + P_2 p^{n-2} + \dots + P_{n-1} p + P_n = 0. \quad \dots (2)$$

Such equations can be solved by one (or more) of some methods given in this chapter. In each of these methods, the given problem is reduced to that of solving one or more equations of the first order and the first degree (already discussed in chapter 2).

Remarks. In this Chapter, we shall present solutions in the following two forms:

(i) The **cartesian form** will consist of x, y and an arbitrary constant c , say. For example, $y^2 = 2cx + c^2$ is general solution in the cartesian form.

(ii) The **parametric form** will consist of two equations of form $x = f_1(p, c)$ and $y = f_2(p, c)$, where c is an arbitrary constant. These two equations together will give the parametric equations of general solution. This situation arises only when the elimination of p is not possible during the process of solving certain equations. Note that p is treated as parameter for this purpose. Sometimes even the above form is not possible. Then we regard the following forms of solutions to be in parametric form: (i) $y = f(x, p), F(x, p, c) = 0$ (ii) $x = g(y, p), G(y, p, c) = 0$.

4.2 Method I. Equations solvable for p

Let $P_0 p^n + P_1 p^{n-1} + P_2 p^{n-2} + \dots + P_{n-1} p + P_n = 0 \quad \dots (1)$ be the given differential equation of the first order and degree $n > 1$.

Since (1) is solvable for p , it can be put in the form

$$[p - f_1(x, y)][p - f_2(x, y)] \dots [p - f_n(x, y)] = 0. \quad \dots (2)$$

Equating each factor of (2) to zero, we obtain n equations of the first order and the first degree, namely,

$$p = dy/dx = f_1(x, y), \quad p = dy/dx = f_2(x, y), \quad \dots, \quad p = dy/dx = f_n(x, y). \quad \dots (3)$$

Let the solutions of these n component equations be respectively

$$F_1(x, y, c_1) = 0, \quad F_2(x, y, c_2) = 0, \quad \dots, \quad F_n(x, y, c_n) = 0,$$

which $c_1, c_2, \dots, c_n$ are the arbitrary constants of integration.

Ex. 9. Solve $p^2 y^2 \cos^2 \alpha - 2pxy \sin^2 \alpha + y^2 - x^2 \sin^2 \alpha = 0$.

Sol. Given

$$(py)^2 - 2(py)x \tan^2 \alpha + (y^2 \sec^2 \alpha - x^2 \tan^2 \alpha) = 0.$$

$$py = \frac{2x \tan^2 \alpha \pm \sqrt{4x^2 \tan^4 \alpha - 4(y^2 \sec^2 \alpha - x^2 \tan^2 \alpha)}}{2}$$

$$py = x \tan^2 \alpha \pm \sqrt{x^2 \tan^2 \alpha (\tan^2 \alpha + 1) - y^2 \sec^2 \alpha}$$

$$y(dy/dx) = x \tan^2 \alpha \pm \sec \alpha \sqrt{x^2 \tan^2 \alpha - y^2}$$

$$y dy - x \tan^2 \alpha dx = \pm \sec \alpha \sqrt{x^2 \tan^2 \alpha - y^2} dx$$

$$\pm \frac{x \tan^2 \alpha dx - y dy}{\sqrt{x^2 \tan^2 \alpha - y^2}} = -\sec \alpha dx.$$

Integrating, $\pm \sqrt{x^2 \tan^2 \alpha - y^2} = c - x \sec \alpha$, c being an arbitrary constant

Squaring, $x^2 \tan^2 \alpha - y^2 = c^2 - 2cx \sec \alpha + x^2 \sec^2 \alpha$ or $x^2 + y^2 - 2cx \sec \alpha + c^2 = 0$.

Ex. 10. Solve $p^2 + 2py \cot x = y^2$.

[Andhra 2003; Srivenkateshwara 2003;

Gorakhpur 2005; Rohilkhand 2009; Kanpur 2008; Gulbarga 2005]

(b) If the curve whose differential equation is $p^2 + 2py \cot x = y^2$ passes through $(\pi/2, 1)$, then the equation of the curve is given by $(2y - \sec^2 x/2)(2y - \operatorname{cosec}^2 x/2) = 0$.

Sol. (a) Given $p^2 + (2y \cot x)p - y^2 = 0$. Solving it for p , we get

$$p = [-2y \cot x \pm \sqrt{(4y^2 \cot^2 x + 4y^2)}]/2$$

$$p = -y \cot x \pm y(\cot^2 x + 1)^{1/2} = -y(\cot x \pm \operatorname{cosec} x).$$

Its component equations are

$$dy/dx = -y(\cot x + \operatorname{cosec} x) \quad \dots (1)$$

$$dy/dx = -y(\cot x - \operatorname{cosec} x). \quad \dots (2)$$

$$(1), \quad \frac{dy}{dx} = -y \left(\frac{\cos x}{\sin x} + \frac{1}{\sin x} \right) = -\frac{1 + \cos x}{\sin x} y = -\frac{2y \cos^2(x/2)}{2 \sin(x/2) \cos(x/2)}$$

$$(1/y) dy = -\cot(x/2) dx.$$

$$\text{Integrating, } \log y = \log c - 2 \log \sin(x/2) \quad \text{or} \quad y = c \operatorname{cosec}^2(x/2). \quad \dots (3)$$

$$\text{By (2), } \frac{dy}{dx} = -y \left(\frac{\cos x}{\sin x} - \frac{1}{\sin x} \right) = \frac{1 - \cos x}{\sin x} y = \frac{2y \sin^2(x/2)}{2 \sin(x/2) \cos(x/2)}$$

$$(1/y) dy = \tan(x/2) dx.$$

$$\text{Integrating, } \log y = \log c - 2 \log \cos(x/2) \quad \text{or} \quad y = c \sec^2(x/2). \quad \dots (4)$$

$\therefore$ From (3) and (4), the combined solution is $(y - c \sec^2 x/2)(y - c \operatorname{cosec}^2 x/2) = 0$.

(b) As in part (a), the general equation of the curve is

$$(y - c \sec^2 x/2)(y - c \operatorname{cosec}^2 x/2) = 0. \quad \dots (5)$$

Since (1) is to pass through $(\pi/2, 1)$, $(5) \Rightarrow (1 - 2c)^2 = 0 \Rightarrow c = 1/2$.

Putting $c = 1/2$ in (5), the equation of the required curve is

$$(2y - \sec^2 x/2)(2y - \operatorname{cosec}^2 x/2) = 0.$$

Ex. 11. Solve $x^2 p^2 - 2xyp + y^2 - x^2 y^2 - x^4 = 0$.

Sol. Given

$$(xp - y)^2 = x^2(x^2 + y^2)$$

or

[Delhi Maths (H) 2004]

$$xp - y = \pm x(x^2 + y^2)^{1/2}$$

or

 dx

or

$$\frac{d(y/x)}{\{1+(y/x)^2\}^{1/2}} = \pm dx$$

Integrating,

$$\text{Thus, } y/x = \sinh(C \pm x)$$

The required solution is

$$\sinh^{-1} t = C \pm x$$

or

or

$$\{y - x \sinh(C+x)\} \{y - x \sinh(C-x)\} = 0$$

Exercise 4(A)

Solve the following differential equations:

1. $p^2 - 5p + 6 = 0$ [Delhi Maths (G) 2005]

2. $p^2 + p = 6$

3. $p^2 - p(e^x + e^{-x}) + 1 = 0$

4. $yp^2 + (x-y)p - x = 0$ [Delhi 2012; Meerut 2005]

5. $x^2 p^2 + 3xyp + 2y^2 = 0$ [Madras 2005, Delhi Maths (G) 1993]

6. $xyp^2 + (3x^2 - 2y^2)p - 6xy = 0$

7. $p^3 + 3xp^2 - y^3 p^2 - 3xy^3 p = 0$

8. $x^2 p^3 + y(1+x^2 y)p^2 + y^3 p = 0$

9. $xyp^2 + (x^2 + xy + y^2)p + x^2 + xy = 0$

10. $x^2 p^2 - 2xyp + y^2 = x^2 y^2 + x^4$

11. $3p^2 y^2 - 2xyp + 4y^2 - x^2 = 0$

12. $y = x \{p + (1+p^2)^{1/2}\}$ [Kanpur 2009]

13. $xyp^2 - (x^2 - y^2)p - xy = 0$ [Madras 2012]

14. $4y^2 p^2 + 2pxy(3x+1) + 3x^3 = 0$ [Delhi 2008]

15. $p^3(x+2y) + 3p^2(x+y) + p(y+2x) = 0$

16. $xyp^2 - (x^2 + y^2)p + xy = 0$ [Gorakhpur 2011]

4.4 Method II. Equations solvable for x

If the given equation $f(x, y, p) = 0$ is solvable for x, we can express x explicitly in terms of y and p. Thus, an equation solvable for x can be put in the form.

$$x = F(y, p) \quad \dots (1)$$

Differentiating (1) w.r.t. 'y' and writing $1/p$ for dx/dy , we get

$$1/p = \phi(y, p, dp/dy), \quad \dots (2)$$

which is a equation involving two variables y and p. Let its solution be

$$\psi(y, p, c) = 0, \quad c \text{ being an arbitrary constant.} \quad \dots (3)$$

Eliminating p between (1) and (3), we get the solution of (1) in the form

$$g(x, y, c) = 0.$$

If the elimination of p between (1) and (3) is not possible, then we solve (1) and (3) to express x and y in terms of p and c in the form

$$x = f_1(p, c),$$

$$y = f_2(p, c). \quad \dots (4)$$

These two equations together form the general solution of (1) in the parametric form, the parameter being p.

Sometimes even the form (4) of the desired solution is not possible. In that case (1) and (3) may be regarded as giving x and y in terms of p, that is, (1) and (3) together are said to form the solution in parametric form.

Remark 1. In some problems (2) can be expressed as $\phi_1(y, p) \phi_2(y, p, dp/dy) = 0$.

In such cases we ignore the first factor $\phi_1(y, p)$ which does not involve dp/dy and proceed with $\phi_2(y, p, dp/dy) = 0$ as already discussed in Art. 4.3

$$(1+t^2)^{1/2}$$

$$\sinh^{-1}(y/x) = C \pm x, \text{ as } t =$$

$$y = x \sinh(C \pm x)$$

$$\{y - x \sinh(C+x)\} \{y - x \sinh(C-x)\} = 0$$

Ans. $(y - 2x - c)(cy - 3x - c)$

Ans. $(y + 3x - c)(y - 2x - c)$

Ans. $(y - x e^2 - c)(y + x e^2 - c)$

Ans. $(y - x - c)(y^2 + x^2 - c)$

Ans. $(xy - c)(x^2 y - c)$

Ans. $(y^2 + 3x^2 - c)(y - cx)$

Ans. $(y - c)(2y + 3x^2 - c)(y^2 + 2x - c)$

Ans. $(y - c)(x - y^{-1} - c)(ye^{-1/x} - c)$

Ans. $(x^2 + y^2 - c)(2xy + x^2 - c)$

Ans. $x + c = \pm \log[(y/x) + \{1 + (y/x)^2\}^{1/2}]$

Ans. $2x \pm (x^2 - 3y^2)^{1/2}$

Ans. $x^2 + y^2 =$

Ans. $(y^2 - x^2 - c)(xy - c)$

Ans. $(x^2 + 2y^2 - c)(x^2 + y^2 - c)$

[Delhi B.Sc. (Prog) II 2010]

Ans. $(y - c)(x + y - c)(xy + x^2 + y^2 - c)$

Ans. $(x^2 - y^2 - c)(y - cx)$

Remark 2. If instead of ignoring the factor $\phi_1(y, p)$, we eliminate p between (1) and $\phi_1(y, p) = 0$, we obtain an equation involving no constant c . This is known as *singular solution* of (1) and we shall discuss it later on in this chapter. Singular solution will be obtained only if asked to do so in a given question. Refer part II of this chapter for more details.

Special case of Method II. Equations that do not contain y .

If the given equation does not contain y , then it can be put in the form $f(x, p) = 0$ (6)

If (6) is solvable for p , it can be put in the form $p = F(x)$ or $dy/dx = F(x)$,

which can be easily integrated to give the required solution.

If (6) is solvable for x , it can be put in the form $x = G(p)$, which is of the form (1) considered in Art 4.3 and so it can be solved as before.

4.5 Solved examples based on method II of Art. 4.4

Ex. 3. Solve the following differential equations :

(a) $y^2 = 2px + y^2 p^3$. [Lucknow 2010; Purvanchal 2010; Delhi B.Sc. (Prog.) II 2008]

(b) $p^3 - 4xyp + 8y^2 = 0$. Meerut 2013; Delhi B.Sc. Maths (Prog.) II 2007]

(c) $(2x + y)p = y - xyp^2$, $a > 0$ [I.A.S. 2001]

Sol. (a) Given $y^2 = 2px + y^2 p^3$ where $p = dy/dx$ (1)

Solving (1) for x , $x = y(1/2p) - y^2(p^2/2)$ (2)

Differentiating (2) w.r.t. 'y' and writing $1/p$ for dx/dy , we get

$$\frac{1}{p} = \frac{1}{2p} - \frac{y}{2p^2} \frac{dp}{dy} - \frac{2y p^2}{2} - \frac{y^2}{2} \cdot 2p \frac{dp}{dy}$$

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$$\frac{1}{2p} + yp^2 + \frac{dp}{dy} \left(\frac{y}{2p^2} + yp^2 \right) = 0 \quad \text{or} \quad p \left(py + \frac{1}{2p^2} \right) + y \frac{dp}{dy} \left(py + \frac{1}{2p^2} \right) = 0$$

$$[py + (1/2 p^2)] [p + y (dp/dy)] = 0.$$

*Neglecting the first factor which does not involve dp/dy , (3) reduces to

$$p + y (dp/dy) = 0 \quad \text{or} \quad (1/p) dp + (1/y) dy = 0.$$

$$\text{Integrating, } \log p + \log y = \log c \quad \text{or} \quad py = c \quad \text{or} \quad p = c/y.$$

Substituting this value of p in (1), we get the required solution

$$y^2 = (2x) (c/y) + y^2 (c/y)^3 \quad \text{or} \quad y^2 = 2cx + c^3.$$

(b) Given $p^3 - 4xyp + 8y^2 = 0$, where $p = dy/dx$ (1)

Solving (1) for x , $x = (1/4y) p^2 + (1/p) (2y)$ (2)

Differentiating (2) w.r.t. 'y' and writing $1/p$ for dx/dy , we get

$$\frac{1}{p} = -\frac{p^2}{4y^2} + \frac{2p}{4y} \frac{dp}{dy} + \frac{2}{p} - \frac{2y}{p^2} \frac{dp}{dy} \quad \text{or} \quad \frac{p^2}{4y} - \frac{1}{p} - \frac{dp}{dy} \left(\frac{p}{2y} - \frac{2y}{p^2} \right) = 0$$

$$\left(\frac{p^2}{4y} - \frac{1}{p} \right) - \frac{2y}{p} \frac{dp}{dy} \left(\frac{p^2}{4y} - \frac{1}{p} \right) = 0 \quad \text{or} \quad \left(\frac{p^2}{4y} - \frac{1}{p} \right) \left(1 - \frac{2y}{p} \frac{dp}{dy} \right) = 0 \quad \dots (3)$$

Neglecting the first factor which does not involve dp/dy , (3) reduces to

$$1 - (2y/p) (dp/dy) = 0 \quad \text{or} \quad (2/p) dp = (1/y) dy.$$

$$\text{Integrating, } 2 \log p = \log y + \log c' \quad \text{so that} \quad p^2 = c' y, \quad \dots (4)$$

We now proceed to eliminate p between (1) and (4). Re-writing (1), $p(p^2 - 4xy) = -8y^2$

* If we take $py + (1/2p^2) = 0$, then we shall get singular solution which is not to be discussed unless asked. Thus, the factor which does not involve a derivative of p w.r.t. 'x' or 'y' will be omitted. For more details refer part II of this chapter.

Differentiating (1) w.r.t. 'y' and writing $1/p$ for dx/dy , we get

$$\frac{1}{p} = 0 - \left[(1+p^2)^{-1/2} + p \cdot \left(-\frac{1}{2}\right) (1+p^2)^{-3/2} \cdot 2p \right] \frac{dp}{dy}$$

$$\frac{1}{p} = \frac{dp}{dy} \left[\frac{p^2}{(1+p^2)^{3/2}} - \frac{1}{(1+p^2)^{3/2}} \right] = \frac{dp}{dy} \frac{p^2 - (1+p^2)}{(1+p^2)^{3/2}}$$

$$dy = -(1+p^2)^{-3/2} p dp.$$

Integrating, $y = c - \frac{1}{2} \int (1+p^2)^{-3/2} (2p dp) = c - \frac{1}{2} \int v^{-3/2} dv$, putting $1+p^2 = v$, $2p dp = dv$

$$y = c + v^{-1/2}$$

or

$$y = c + (1+p^2)^{-1/2}. \quad \dots (2)$$

We now try to eliminate p from (1) and (2) as follows.

$$\text{Here, (1) and (2)} \Rightarrow x - a = p/(1+p^2)^{1/2} \quad \text{and} \quad y - c = 1/(1+p^2)^{1/2}.$$

$$\text{Squaring and adding these,} \quad (x-a)^2 + (y-c)^2 = 1,$$

which is the required solution, c being an arbitrary constant.

Exercise 4(B)

Solve the following differential equations

1. $p^3 y^2 - 2px + y = 0$ [Delhi B.Sc. (Prog) II 2011]

Ans. $y^2 = 2cx - c^3$

2. $yp^2 - 2xp + y = 0$ [Lucknow 2008; Meerut 2005; Purvanchal 2009]

Ans. $y^2 = 2cx - c^2$

3. (i) $y - 2xp + ayp^2 = 0$ (ii) $y - 2xp + yp^2 = 0$ (I.A.S. 2008)

Ans. (i) $y^2 = 2cx - ac^2$ (ii) $y = 2cx - c^2$

4. $4(xp^2 + yp) = y^4$

Ans. $y = 4c(xy + 1)$

5. $y = 3px + 6p^2 y^2$ [Delhi Maths (H) 2001; Delhi Maths (G) 2000]

Ans. $y^3 = 3cx + 6c^2$

6. $p^2 - 2xp + 1 = 0$

Ans. $x = (p/2) + (1/2p)$, $y = (p^2/4) - (1/2) \times \log p + c$

7. $xp^3 = a + bp$

Ans. $x = a/p^3 + b/p^2$, $y = (3a/2p^2) + (2b/p) + c$

8. $xp^3 - p^2 - 1 = 0$

Ans. $x = p^{-1} + p^{-3}$, $y = (3/2) \times p^{-2} - \log p + c$

9. $p = \tan \{x - p/(1+p^2)\}$ or $p \cot \{x - p/(1+p^2)\} = 1$

Ans. $x = \{p/(1+p^2)\} + \tan^{-1} p$, $y = c - (p^2 + 1)^{-1}$

1.6 Method III. Equations solvable for y

If the given equation $f(x, y, p) = 0$ is solvable for y , we can express y explicitly in terms of x and p . Thus, an equation solvable for y can be put in the form $y = F(x, p)$ (1)

Differentiating (1) w.r.t. x and writing p for dy/dx , we get $p = \phi(x, p, dp/dx)$, ... (2)

which is an equation involving two variables x and p . Let its solution be $\psi(x, p, c) = 0$, c being an arbitrary constant. ... (3)

Eliminating p between (1) and (3), the solution of (1) is in the form $g(x, y, c) = 0$.

If the elimination of p between (1) and (3) is not possible, then we solve (1) and (3) to express x and y in terms of p and c in the form

$$x = f_1(p, c), \quad y = f_2(p, c). \quad \dots (4)$$

These two equations together form the general solution of (1) in the parametric form, the parameter being p .

Sometimes even the form (4) of the desired solution is not possible. In that case (1) and (3) may be regarded as giving x and y in terms of p , that is, (1) and (3) together are said to form the solution in parametric form.

Remark 1. In some problems (2) can be expressed as $\phi_1(x, p) \phi_2(x, p, dp/dx) = 0$ (5)

In such cases we ignore the first factor $\phi_1(x, p)$ which does not involve dp/dx and proceed with $\phi_2(x, p, dp/dx) = 0$ as discussed in Art. 4.6.

Remark 2. If instead of ignoring the factor $\phi_1(x, p)$, we eliminate p between (1) and $\phi_1(x, p) = 0$, we obtain an equation involving no constant c . This is known as *singular solution* (1) and we shall discuss it later on in this chapter. Singular solution will be obtained only if asked to do so in a given question. For details refer part II of this chapter.

First special case of Method III. Lagrange's equation

The differential equation of the form $y = x F(p) + f(p)$ is known as *Lagrange's equation*. To solve it we proceed as in Art. 4.6.

Differentiating (6) w.r.t. 'x', we have

$$\text{or } p - F(p) = \frac{dp}{dx} [x F'(p) + f'(p)]$$

$$\text{or } \frac{dx}{dp} - \frac{F'(p)}{p - F(p)} x = \frac{f'(p)}{p - F(p)},$$

which is linear equation in x and p and can be solved by usual method to give a relation of the form $x = \phi(p, c)$

We now eliminate p between (6) and (7) to get the required solution

If p cannot be eliminated, then putting the value of x in (6), we get

$$y = \phi(p, c) F(p) + f(p).$$

Then (7) and (8) together form the required solution in parametric form, p being regarded parameter.

Second special case of method III. Equation that do not contain x .

If given equation does not contain x , then it can be put in the form $f(y, p) = 0$.

If (9) is solvable for p , it can be put in the form $p = F(y)$ or $dy/dx = F(y)$

which can be easily integrated to give the required solution.

If (9) is solvable for y , it can be put in the form

$$y = G(p),$$

which is of the form (1) of Art. 4.6 and so it can be solved as before.

4.7 Solved examples based on method III of Art. 4.6.

Ex. 1. Solve (a) $y = 3x + \log p$.

(b) $y = x \{p + (1 + p^2)^{1/2}\}$.

[Rohilkhand 2009; Kanpur 2009]

Sol. (a) Given

$$y = 3x + \log p, \text{ where } p = dy/dx.$$

Differentiating (1) w.r.t. 'x' and writing p for dy/dx , we get

$$p = 3 + (1/p) (dp/dx)$$

or

$$p(p - 3) = (dp/dx)$$

or

$$\frac{dx}{dp} = \frac{dp}{p(p - 3)} = \frac{1}{3} \left[\frac{1}{p - 3} - \frac{1}{p} \right] dp, \text{ resolving into partial fractions.}$$

Integrating,

$$x = (1/3) [\log(p - 3) - \log p] - (1/3) \log c, \text{ } c \text{ being an arbitrary constant}$$

or

$$\log \left[\frac{p - 3}{pc} \right] = 3x$$

or

$$\frac{p - 3}{pc} = e^{3x}$$

or

$$p = \frac{3}{1 - ce^{3x}}$$

Putting this value of p in (1), the required solution is

$$y = 3x + \log \{3/(1 - ce^{3x})\}, \text{ } c \text{ being an arbitrary constant}$$

(b) Given

$$y = x \{p + (1 + p^2)^{1/2}\}, \text{ where } p = dy/dx.$$

Differentiating (1) w.r.t. 'x' and writing p for dy/dx , we get

$$p = p + (1 + p^2)^{1/2} + x \left[1 + \frac{1}{2}(1 + p^2)^{-1/2} \times 2p \right] \frac{dp}{dx}$$

$$\text{or } \frac{x[p + (1 + p^2)^{1/2}]}{(1 + p^2)^{1/2}} \frac{dp}{dx} = -(1 + p^2)^{1/2} \quad \text{or} \quad \left[\frac{1}{2} \cdot \frac{2p}{1 + p^2} + \frac{1}{(1 + p^2)^{1/2}} \right] dp = -\frac{dx}{x}$$

$$\text{Integrating, } (1/2) \times \log(1 + p^2) + \log[p + (1 + p^2)^{1/2}] = \log c - \log x$$

$$\log(c/x) = \log(1 + p^2)^{1/2} + \log[p + (1 + p^2)^{1/2}]$$

$$\text{or } c/x = (1 + p^2)^{1/2} [p + (1 + p^2)^{1/2}] \quad \text{or } c/x = (1 + p^2)^{1/2} (y/x), \text{ by (1)}$$

$$\text{or } c = y(1 + p^2)^{1/2} \quad \text{or } c^2 = y^2(1 + p^2) \quad \text{or } c^2 - y^2 = y^2 p^2 \quad \dots (2)$$

$$\text{Multiplying both sides of (1) by } y, \quad y^2 = xyp + xy(1 + p^2)^{1/2}$$

$$\text{or } y^2 = xyp + cx, \text{ since from (2)} \quad y(1 + p^2)^{1/2} = c$$

$$\text{or } y^2 - cx = xyp \quad \text{or } (y^2 - cx)^2 = x^2 y^2 p^2$$

$$\text{or } y^4 - 2cxy^2 + c^2 x^2 = x^2 y^2 p^2 \quad \text{or } y^4 - 2cxy^2 + c^2 x^2 = x^2 (c^2 - y^2), \text{ by (2)}$$

$$\text{or } y^4 - 2cxy^2 + x^2 y^2 = 0 \quad \text{or } x^2 + y^2 - 2xc = 0$$

Ex. 2. Solve the following differential equations:

(a) $y + px = x^4 p^2$. [M.S. Univ. T.N. 2007; Agra 2009, Avadh 2008; Lucknow 2011; Delhi B.A. (Prog.) II 2013; Rohilkhand 2015; Purvanchal 2011]

(b) $y = yp^2 + 2px$ [Allahabad 1994, Kanpur, 1993]

(c) $y = 2px + f(xp^2)$ Meerut 2007, Delhi Maths (G) 1998

(d) $y = 2px + \tan^{-1}(xp^2)$ [Nagarjuna 2003]

Sol. (a) Given $y + px = x^4 p^2$, where $p = dy/dx$ (1)

Solving (1) for y , $y = x^4 p^2 - px$ (2)

Differentiating (2) w.r.t. 'x' and writing p for dy/dx , we get

$$p = 4x^3 p^2 + 2x^4 p (dp/dx) - [p + x(dp/dx)]$$

$$\text{or } 2p - 4x^3 p^2 + (dp/dx)(x - 2x^4 p) = 0 \quad \text{or } 2p(1 - 2x^3 p) + x(dp/dx)(1 - 2x^3 p) = 0$$

$$\text{or } (1 - 2x^3 p)[2p + x(dp/dx)] = 0. \quad \dots (3)$$

Neglecting the first factor which does not involve dp/dx , (3) reduces to

$$2p + x(dp/dx) = 0 \quad \text{or } (1/p) dp + 2(1/x) dx = 0.$$

$$\text{Integrating, } \log p + 2 \log x = \log c \quad \text{or } px^2 = c \quad \text{or } p = c/x^2.$$

Putting this value of p in (1), the required solution is

$$y + x(c/x^2) = x^4 (c^2/x^4) \quad \text{or } xy + c = c^2 x. \quad \dots (1)$$

(b) Given $y = yp^2 + 2px$, where $p = dy/dx$.

Solving (1) for y , $y(1 - p^2) = 2px$ or $y = 2px/(1 - p^2). \quad \dots (2)$

Differentiating (2) w.r.t. 'x' and writing p for dy/dx , we get

$$p = [(1 - p^2) \{2p + 2x(dp/dx)\} - 2px(-2p)(dp/dx)] / (1 - p^2)^2$$

$$p(1 - p^2)^2 = 2p(1 - p^2) + 2x(1 - p^2)(dp/dx) + 4p^2 x(dp/dx)$$

$$p(1 - p^2)[(1 - p^2) - 2] - 2x(dp/dx)(1 - p^2 + 2p^2) = 0$$

$$p(p^2 - 1)(1 + p^2) - 2x(dp/dx)(1 + p^2) = 0 \quad \dots (3)$$

$$(1 + p^2)[p(p^2 - 1) - 2x(dp/dx)] = 0.$$

Neglecting the first factor which does not involve dp/dx , (3) reduces to

$$p(p^2 - 1) - 2x \frac{dp}{dx} = 0 \quad \text{or} \quad \left(\frac{1}{p-1} + \frac{1}{p+1} - \frac{2}{p} \right) dp = \frac{dx}{x}.$$